

JEE MAIN REVISION MASTER: TRIGONOMETRIC RATIO AND IDENTITIES

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 69 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Trigonometric Ratio And Identities - Core Application	69 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Trigonometric Ratio And Identities - Core Application (Very High Yield — 69 PYQs)

Core Principle: Key problem pattern in Trigonometric Ratio And Identities.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Let $\tan A, \tan B$, where $A, B \in (-\pi/2, \pi/2)$, be the roots of the quadratic equation $x^2 - 2x - 5 = 0$. Then $\sin^2 \{ \dots \}$ " [Ans: C]
- [2026 Morning Shift] Q#1 (MCQ): "The value of $\frac{\sqrt{3} \operatorname{cosec} 20^\circ - \sec 20^\circ \cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ}{\dots}$ is equal to..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (Numerical): "If $A = \frac{\sin 3^\circ \cos 9^\circ + \sin 9^\circ \cos 27^\circ + \sin 27^\circ \cos 81^\circ}{\dots}$ and $B = \tan 81^\circ - \tan 3^\circ$ " [Ans: 2]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 5th April Morning Shift (Q#1)

Let $\tan A, \tan B$, where $A, B \in (-\pi/2, \pi/2)$, be the roots of the quadratic equation $x^2 - 2x - 5 = 0$. Then $20\sin^2((A+B)/2)$ is equal to:

- (A) $10 + \sqrt{10}$
- (B) $10 - 2\sqrt{10}$
- (C) $10 - 3\sqrt{10}$ [Correct]
- (D) $10 - \sqrt{10}$

Correct Answer: C

Solution Analysis: Let the roots of the equation $x^2 - 2x - 5 = 0$ be $\tan A$ and $\tan B$. So, by using relations between roots and coefficients, $\tan A + \tan B = 2$ and $\tan A \tan B = -5$. We have to find $20\sin^2((A+B)/2)$. First, find $\tan(A+B)$ using the formula $\tan(A+B) = (\tan A + \tan B) / (1 - \tan A \tan B)$. Substituting the values: $\tan(A+B) = (2 / (1 - (-5))) = (2 / 6) = (1 / 3)$. Now let $\theta = A+B$. Then $\tan \theta = (1 / 3)$. We need $\sin^2(\theta/2)$. Using the identity $\sin^2(\theta/2) = (1 - \cos \theta) / 2$, we first find $\cos \theta$. Since $\tan \theta = (1 / 3)$, take a right triangle with